

Eigenvalues and eigenvectors

Definition

Suppose that A is an $n \times n$ matrix and $\lambda \in \mathbb{R}$.

- 1 If $u \in \mathbb{R}^n$ with $u \neq 0$ and

$$Au = \lambda u$$

then we say that λ is an eigenvalue of A and u is an eigenvector for λ .

- 2 If λ is an eigenvalue for A then

$$\{u \in \mathbb{R}^n : \underline{Au = \lambda u}\} = \text{Nul}(A - \lambda I)$$

is called the λ -eigenspace for A .


$$\underline{Au} = \underline{\lambda I}u$$
$$\underline{(A - \lambda I)}u = 0$$

Linear independence of eigenvectors

Theorem

Suppose that A is an $n \times n$ matrix and $v_1, \dots, v_k$ are eigenvectors for distinct eigenvalues $\lambda_1, \dots, \lambda_k$. Then $v_1, \dots, v_k$ are linearly independent.

$$* \rightarrow c_1 v_1 + c_2 v_2 + c_3 v_3 = 0 \quad c_1, c_2, c_3 \neq 0$$

Mult.
by A

$$c_1 A v_1 + c_2 A v_2 + c_3 A v_3 = 0$$

$$A v_1 = \lambda_1 v_1$$

$$A v_2 = \lambda_2 v_2$$

$$A v_3 = \lambda_3 v_3$$

$$\cancel{c_1 \lambda_1} v_1 + c_2 \lambda_2 v_2 + c_3 \lambda_3 v_3 = 0$$

Mult *
by λ_1

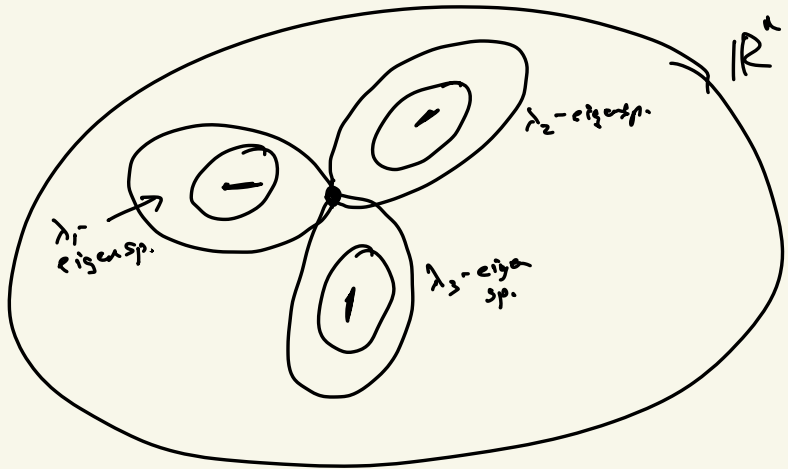
$$\cancel{c_1 \lambda_1} v_1 + c_2 \lambda_1 v_2 + c_3 \lambda_1 v_3 = 0$$

$$\lambda_1 \neq \lambda_2 \neq \lambda_3$$

$$\underbrace{c_2 (\lambda_2 - \lambda_1)}_{\neq 0} v_2 + \underbrace{c_3 (\lambda_3 - \lambda_1)}_{\neq 0} v_3 = 0$$

~~✗~~.

Fix A , $n \times n$ matrix with eigenvalues $\lambda_1, \dots, \lambda_k$.



The characteristic equation

Definition

For A , an $n \times n$ matrix, $\det(A - \lambda I_n) = 0$ is called the characteristic equation of A . It is a polynomial of degree n .



- 1 The eigenvalues of A are solutions of the characteristic equation of A .
- 2 The algebraic multiplicity of an eigenvalue λ is the multiplicity of λ as a root of the characteristic equation.
- 3 The geometric multiplicity of λ is the dimension of the λ -eigenspace.
- 4 The geometric multiplicity is always less than or equal to the algebraic multiplicity.

→ A consequence is that for any $n \times n$ matrix A there are at most n eigenvalues.

The fact that geom mult (ie dim of the eigenspace)

is $\leq$ alg mult. is really helpful when

the alg. mult. is 1.

Ex:

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Eigenvalues?

$$\det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 1 & 0 \\ 1 & 1-\lambda & 0 \\ 0 & 0 & 2-\lambda \end{pmatrix}$$

$$= (2-\lambda)((1-\lambda)^2 - 1)$$

$$= (2-\lambda)(\lambda^2 - 2\lambda)$$

$$= \frac{\lambda(2-\lambda)(\lambda-2)}{\lambda(2-\lambda)(\lambda-2)} \leftarrow \begin{matrix} \text{deg. 3} \\ \text{poly.} \end{matrix}$$

eigenvalues are $\lambda = 0$ or 2 .

alg. mult. is 1.

alg. mult. of 2 is 2

Eigenvectors:

$$\lambda = 0 : \text{ Solve } (A - \lambda I)u = 0$$

$$\text{Solve } \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$u_3 = 0 \quad \text{and} \quad \underline{u_1 + u_2 = 0}$$

$$u = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} t \quad \text{for } t \in \mathbb{R}.$$

The eigenspace for $\lambda = 0$ has dim. 1.

$\lambda = 2$: Solve $(A - 2I)u = 0$

$$\begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ \underline{u_3} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$u_3 = t$ for any $t \in \mathbb{R}$. and $u_1 = u_2$

$$u = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} s + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} t \quad \text{for } s, t \in \mathbb{R}.$$

The 2-eigenspace has
dim 2. = the alg.
mult. of $\lambda = 2$.

Ex: $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ Eigenvalues?

$$\det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 1 \\ 0 & 1-\lambda \end{pmatrix}$$

$$= (1-\lambda)^2 \quad \text{so } \lambda = 1 \text{ is the only eigenvalue.}$$

with alg. mult 2.

Eigenvectors: $\lambda = 1$

Solve $(A - \lambda I)u = 0$ $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$u_1 = t, u_2 = 0$ so $u = t \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ for $t \in \mathbb{R}$.

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} u_2 &= 0 \\ 0 &= 0 \end{aligned}$$

$$u = \begin{pmatrix} t \\ 0 \end{pmatrix} \text{ for } t \in \mathbb{R}.$$

is the solution.

The 1-eigenspace has dim 1 !

But $\lambda = 1$ has alg. mult. 2.

Similarity

Definition

For A and B , two $n \times n$ matrices, we say that A is similar to B if there is an invertible matrix P such that $P^{-1}AP = B$.

Fact: If A and B are similar then they have the same characteristic polynomial.